Existing Quantum Error-Correction Methods: Comparative Study and a Proposed Best-Fit Architecture for Scalable Fault Tolerance

Research Paper

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Abstract

Quantum computing is limited not only by gate infidelity and decoherence, but by the difficulty of preserving fragile quantum information long enough to execute useful algorithms. Quantum error correction (QEC) is therefore the central engineering problem between today's noisy devices and large-scale fault-tolerant quantum computation. This paper presents a comparative study of the main families of existing QEC methods: concatenated Calderbank-Shor-Steane codes, surface codes, color codes, bosonic and Gottesman-Kitaev-Preskill (GKP) codes, and quantum low-density parity-check (qLDPC) codes. The analysis is literature-based rather than experimental; it synthesizes foundational theory, representative decoder research, and recent hardware milestones from superconducting, neutral-atom, and bosonic platforms. The comparison is organized around six criteria: threshold or error tolerance, physical-qubit overhead, geometric locality, decoder latency and complexity, logical-gate friendliness, and experimental maturity. The main conclusion is that no single code family is best in every regime. However, for near-term general-purpose fault-tolerant computation, the strongest current choice is a surface-code-centric architecture enhanced by leakage removal, real-time intelligent decoding, and modular lattice-surgery operations. qLDPC codes appear most promising for long-term overhead reduction, while GKP-style bosonic encodings are highly attractive for hardware platforms that can generate and stabilize oscillator states. On the basis of this comparison, the paper proposes an Adaptive Hybrid Surface Architecture: a best-fit current strategy that keeps the experimentally mature surface code as the computational backbone, augments it with fast classical decoding and learned reranking, and preserves a migration path toward qLDPC memories and bosonic front-end encodings as those technologies mature.

Keywords: quantum error correction, surface code, color code, GKP code, qLDPC, fault tolerance, logical qubits, decoder design

1. Introduction

Quantum computing promises computational advantages in chemistry, optimization, materials discovery, and cryptanalysis, but physical quantum devices remain fragile. Decoherence, imperfect control pulses, crosstalk, measurement faults, state-preparation errors, and leakage outside the computational basis all degrade computation long before large algorithms can finish. The central response to this challenge is quantum error correction (QEC), which encodes one logical qubit into a larger set of physical degrees of freedom so that errors can be detected and corrected without destroying the encoded information [1]-[6].

The field of QEC began with the earliest qubit codes of Shor and Steane, which showed that the apparent impossibility of protecting unknown quantum states could be overcome by structured redundancy and syndrome extraction [1], [2]. The topological turn led by Kitaev and by Dennis, Kitaev, Landahl, and Preskill reframed error correction as a problem of geometry, locality, and phase transitions [3], [4]. Later work by Fowler and collaborators made the surface code the leading candidate for practical large-scale architectures because of its high threshold, locality, and decoder compatibility [6]. In parallel, researchers developed color codes, bosonic encodings such as GKP, and increasingly sophisticated qLDPC code constructions [8], [9], [13]-[15].

The last few years have transformed QEC from a mostly theoretical subject into an experimental race. Superconducting processors have demonstrated repeated surface-code cycles, break-even logical behavior, leakage suppression, advanced learned decoders, and below-threshold logical memories [10]-[12], [17], [19]. Neutral-atom systems have demonstrated logical processors with many encoded qubits

and impressive logical operations across multiple code families [16]. On the theory side, IBM and others have shown that qLDPC memories may drastically reduce physical-qubit overhead while preserving useful thresholds [13]-[15]. These milestones make it timely to ask a focused engineering question: among the existing approaches, which one is best today, and what should a research-driven near-term architecture look like?

This paper answers that question in a careful, qualified sense. It does not claim a universal winner for all platforms or asymptotic regimes. Instead, it defines best as the strongest overall choice for near-term scalable, general-purpose fault-tolerant quantum computing when experimental maturity, decoder practicality, and hardware constraints matter as much as asymptotic code distance. The central thesis is that the best current choice is not a single isolated code family, but a surface-code-centric hybrid architecture that integrates the strongest experimentally validated features already visible in the literature. qLDPC and bosonic encodings are not dismissed; they are treated as crucial future layers in a staged architecture rather than immediate full replacements.

This paper makes four contributions. First, it organizes the existing QEC landscape into major code families and places them in a common decision framework. Second, it compares those families using qualitative and semi-quantitative criteria that matter for implementation rather than only for asymptotic theory. Third, it synthesizes recent experimental milestones from the major hardware platforms driving the field. Fourth, it proposes a best-fit current architecture that is realistic enough to guide simulation-based student research and sufficiently forward-looking to connect with the problems being pursued at laboratories such as Google Quantum AI, IBM, and leading university groups.

2. Method of Comparative Review

This paper is a comparative review rather than a laboratory report or a strict statistical meta-analysis. The literature corpus was intentionally limited to representative papers that shape real architectural choices. Twenty core papers from 1995 to 2025 were used to anchor the comparison, with additional sources used for contextual clarification. The corpus was balanced across five categories: foundational code theory, practical surface-code architecture, alternative code families, decoder research, and recent experimental demonstrations on major hardware platforms. This design reflects the fact that a useful QEC paper must compare mathematics, control systems, and hardware evidence at the same time.

The selection logic followed four filters. First, a paper was included if it established a foundational concept still used in contemporary QEC, such as stabilizer measurement, topological protection, or bosonic encoding. Second, a paper was included if it changed the practical evaluation of a code family by reporting a hardware milestone, such as repeated QEC cycles, break-even performance, below-threshold memory, or large logical processors. Third, decoder papers were included when they changed the real-time feasibility of a code family rather than merely improving a theoretical bound. Fourth, the set was diversified across superconducting, neutral-atom, and bosonic settings so that the paper would not silently equate one hardware platform with the entire field.

The review is therefore intentionally comparative and engineering-focused. Papers were not ranked only by citation count or novelty. Instead, they were weighted by how much they changed the answer to the design question posed in this paper: what should a researcher pick as the best current architecture if the goal is scalable, general-purpose fault tolerance rather than a narrow platform-specific victory? This

method also explains why the final recommendation is hybrid. Once literature is compared through an implementation lens, it becomes clear that different code families win different subproblems.

Study-design filters

- Foundational influence: the paper introduced a concept still shaping current QEC design.
- Experimental significance: the paper demonstrated a credible milestone on real hardware.
- Decoder relevance: the paper changed practical classical processing requirements.
- Hardware diversity: the corpus should not be dominated by one platform or one research group.

3. Background: Why Quantum Error Correction Is Hard

Classical error correction relies on copying bits and comparing them. Quantum information cannot be protected that way because unknown quantum states cannot simply be copied and repeatedly observed without disturbance. QEC therefore works indirectly. Instead of measuring the logical state itself, one measures stabilizers: operators whose collective outcomes, called syndromes, reveal the presence and likely location of errors while preserving the encoded information [2], [5]. A logical qubit is thus not a single physical object; it is a protected subspace distributed across many physical qubits or modes.

Fault tolerance extends this idea. A code is not enough if the correction procedure itself spreads errors faster than the code removes them. Fault-tolerant protocols therefore constrain how syndrome measurements, ancilla preparation, resets, and logical operations are performed. In practice, the key concept is the threshold: below a certain effective physical error rate, increasing the code distance reduces the logical error rate instead of making matters worse [4]-[6]. For topological codes, one commonly writes a heuristic scaling relation of the form $P_L \approx A (p/p_{th})^{((d+1)/2)}$, where P_L is the logical error rate, p is the physical error rate, p_{th} is the threshold, and d is the code distance. The equation is not universal across all noise models, but it captures the basic idea that logical performance improves exponentially with distance only in the below-threshold regime.

Because the literature reports heterogeneous metrics, this paper uses a practical evaluation framework instead of pretending that every code family can be compared by one number. Six criteria are emphasized. Threshold or error tolerance matters because it determines how good the physical hardware must be before scaling helps. Physical-qubit overhead matters because extremely low logical error rates may require thousands or millions of physical components. Geometric locality matters because two-dimensional nearest-neighbor devices are easier to build than architectures with global couplings. Decoder complexity matters because syndrome information must be processed in real time by a classical coprocessor. Logical-gate friendliness matters because some codes make Clifford or non-Clifford operations much cheaper than others. Experimental maturity matters because a beautiful asymptotic theory is not yet the same as a reliable hardware path [5], [6], [13]-[15].

To keep the comparison honest, the study treats code families as engineering options rather than abstract theorems. The best code on paper may be unusable on a given chip because its stabilizers are too high-weight, its required connectivity is unrealistic, or its decoder is too slow. Conversely, a code with worse asymptotic overhead may still be the best present choice if it is compatible with existing qubit layouts, ancilla reuse, calibration pipelines, and real-time classical hardware. This implementation-first viewpoint is one reason the surface code remains dominant in experiments even while qLDPC research accelerates.

Criterion	Why it matters	Preferred direction
Threshold / tolerance	Determines whether scaling the code reduces logical errors on realistic hardware.	Higher
Physical-qubit overhead	Controls cost per logical qubit and total system size for useful algorithms.	Lower
Geometric locality	Favors layouts that fit planar or modular hardware with limited couplers.	More local
Decoder latency / complexity	Syndromes must be processed fast enough to keep up with correction cycles.	Lower / faster
Logical-gate friendliness	Some codes make Clifford or non-Clifford gates much cheaper.	Higher
Experimental maturity	Hardware evidence lowers architectural risk and shortens the path to scaling.	Higher

Table 1. Evaluation criteria used throughout the comparison.

4. Existing Families of Quantum Error Correction

4.1 Concatenated and CSS Code Families

The earliest practical QEC schemes were concatenated block codes, especially the Shor and Steane constructions [1], [2]. Their historical importance is enormous: they proved that quantum states could be protected without direct measurement of the encoded data, and they established the stabilizer framework that still underlies most modern QEC [2], [5]. CSS-style constructions also exposed a valuable design feature: some logical gates can be implemented transversally or with relatively clean algebraic structure, which makes them attractive for reasoning about fault-tolerant gate sets.

Despite these strengths, concatenated block codes have become less central for large-scale nearest-neighbor architectures. Their syndrome extraction often requires nontrivial routing or more global connectivity than planar chips naturally provide, and under local hardware constraints they generally lose their implementation advantage to topological codes [5], [6]. They are still important conceptually and remain relevant in some modular or ion-trap settings, but they no longer define the main experimental frontier for very large systems. In this study, they are best understood as foundational codes and as useful reference points for gate design, not as the strongest default choice for general-purpose near-term scaling.

4.2 Surface Codes

Surface codes dominate the current practical discussion of QEC because they balance threshold, locality, and decoder ecosystem unusually well [4], [6]. In the surface code, qubits are arranged on a two-dimensional lattice, stabilizers have local support, and logical operators are represented by nontrivial paths across the lattice. This geometric structure is hardware-friendly for superconducting and other planar platforms, and it allows scalable operations via techniques such as lattice surgery rather than highly nonlocal gates [7].

The surface code’s threshold under standard circuit-level assumptions is often quoted at around 1%, which is unusually favorable among practical code families [4], [6], [15]. Just as important, the decoder ecosystem is rich. Maximum-likelihood and tensor-network-inspired decoders provide strong accuracy, while union-find and related methods offer much faster nearly linear-time performance [17], [22], [23]. That combination of high threshold and mature decoding has made the surface code the main experimental benchmark for whether QEC is 'working' on real hardware.

Recent milestones reinforce that position. Krinner and colleagues demonstrated repeated QEC in a distance-three surface code with a fast 1.1 microsecond cycle on a superconducting device [10]. Google Quantum AI then showed in 2023 that increasing code distance from $d = 3$ to $d = 5$ could modestly improve logical behavior, an early break-even scaling milestone [11]. Miao and collaborators showed that explicit leakage removal can reduce steady-state leakage by roughly an order of magnitude, which is crucial because leakage breaks the assumptions underlying many decoders [12]. In 2025, Google reported below-threshold surface-code memories on the Willow processor, including a 101-qubit distance-seven code with a reported 0.143% logical error per correction cycle and a suppression factor $\Lambda = 2.14 \pm 0.02$ when increasing distance by two [19].

The main weakness of the surface code is overhead. The code is space hungry, and the cost of universal computation remains strongly influenced by magic-state distillation and the area-time cost of logical patches [6], [7]. In other words, the surface code is easy to believe in but expensive to scale. That fact motivates interest in color codes, bosonic encodings, and qLDPC alternatives.

4.3 Color Codes

Color codes are another topological family, but they differ from surface codes in a strategically important way: they support more efficient logical gate implementations, especially transversal Clifford operations [8]. This makes them attractive for architectures in which logical gate cost matters as much as memory protection. Historically, color codes were compelling on paper yet difficult to realize at scale because their syndrome extraction and decoding are more complicated than those of the surface code.

Recent experiments have made color codes far more relevant. Bluvstein and collaborators demonstrated a logical quantum processor on reconfigurable neutral-atom arrays with up to 280 physical qubits and showed that logical encoding could improve algorithmic performance; their system employed multiple encodings, including color-code qubits with break-even behavior [16]. In 2025, Lacroix and collaborators demonstrated scaling and logic in the color code on a superconducting quantum processor, showing suppression when scaling from distance three to five and logical operations such as transversal Clifford gates, magic-state injection above 99% with post-selection, and lattice-surgery teleportation [20].

These results matter because they challenge a common simplification: that the surface code is the only realistic topological code for superconducting hardware. The color code now appears to be a serious contender whenever one values logical gate efficiency and not just memory threshold. Still, the color code is not yet the best default architecture. Its decoder and stabilizer-measurement complexity remain a larger systems burden than for the surface code, and its experimental maturity is newer and thinner. Thus, in the present comparison, the color code is a high-potential but not yet fully de-risked option.

4.4 Bosonic and GKP Codes

Bosonic codes protect information by embedding a qubit into an oscillator rather than into a collection of two-level systems. The GKP proposal by Gottesman, Kitaev, and Preskill is the most influential example:

it encodes a logical qubit in grid-like states of position and momentum, allowing small shifts in phase space to be corrected [9]. Bosonic codes are attractive because they can compress some redundancy into a single high-quality mode, thereby changing the resource tradeoff from many qubits to fewer but more specialized physical subsystems.

From a systems perspective, bosonic QEC is a hardware-specific superpower rather than a universal default. On platforms that support cavity or oscillator modes with long lifetimes, bosonic encodings can be remarkably efficient. Noh and Chamberland showed that concatenating GKP with the surface code can achieve fault tolerance under realistic assumptions, making the surface-GKP stack a credible route to lower overhead if sufficiently good GKP states can be prepared [21]. In 2024, Lachance-Quirion and collaborators experimentally demonstrated autonomous QEC of GKP states in a superconducting device, showing a logical-lifetime increase that crossed the key threshold of correcting more errors than were introduced [18].

The limitation is obvious: GKP and related bosonic codes are not drop-in solutions for every hardware platform. They require oscillator engineering, state preparation, and often demanding control of squeezing or reservoir engineering. For that reason, bosonic codes rank very highly in hardware-specific efficiency but lower in cross-platform generality. In this paper's proposal, they appear not as the sole backbone for all systems, but as optional front-end encodings that can reduce the effective error burden before a qubit-level topological code takes over.

4.5 Quantum Low-Density Parity-Check (qLDPC) Codes

qLDPC codes are the most exciting asymptotic alternative to the surface code. Their core promise is simple: reduce overhead by maintaining sparse checks while improving encoding rate and distance scaling relative to traditional topological codes [13], [14]. Breuckmann and Eberhardt's perspective helped organize the field and clarify why qLDPC codes may eventually outperform the surface code on total resource cost [13]. Their balanced-product construction provided one explicit nonrandom family with strong scaling properties [14].

The strongest recent milestone came from Bravyi and collaborators, who presented a high-threshold low-overhead fault-tolerant memory based on a family of qLDPC codes and reported an error threshold of about 0.8% under a standard circuit-level noise model [15]. Their paper also emphasized the enormous resource implications: preserving 12 logical qubits for nearly a million syndrome cycles at a 0.1% physical error rate required 288 physical qubits in their example, whereas a comparable surface-code memory would require roughly an order of magnitude more qubits [15]. This is exactly the kind of result that keeps the surface code from being the inevitable long-term winner.

Yet qLDPC is not automatically the best current choice. Sparse parity-check matrices do not by themselves solve implementation problems. The hardware graph can be more demanding, syndrome extraction must be carefully co-designed with the layout, and decoders must still operate at scale with realistic noise including measurement faults and correlated events. In short, qLDPC codes are probably the most important long-term route to overhead reduction, but their full hardware stack is less experimentally mature than the surface-code stack. In this study they rank as the most promising medium- to long-term architecture for memory-heavy systems, but not yet as the most de-risked full-stack choice for near-term general-purpose computing.

5. Comparative Analysis of the Existing Methods

To operationalize the comparison, Table 2 scores the main code families across the six criteria introduced earlier. The purpose is not to collapse all QEC into one simplistic ranking, but to make the design tradeoffs explicit.

Family	Thresh.	Overh.	Local.	Decode	Logic	Maturity
Concatenated CSS	2	2	2	3	4	3
Surface code	5	2	5	5	3	5
Color code	3	3	4	3	5	4
Bosonic / GKP	4*	4*	3	3	4	3
qLDPC	4	5	2	2	3	2

Table 2. Qualitative score matrix (1 = weakest, 5 = strongest). Bosonic / GKP scores are marked with an asterisk because they are highly platform dependent.

Table 2 offers a qualitative synthesis rather than an absolute scoreboard. Scores range from 1 (weakest) to 5 (strongest), and they reflect the literature as a whole, not a single lab benchmark. The surface code earns the highest total practical score because it combines strong threshold behavior, planar locality, mature decoders, and the deepest experimental validation [4], [6], [10]-[12], [17], [19]. Its weak point is overhead. That weakness is serious, but it is also visible and manageable, which is one reason engineers prefer it over less mature alternatives.

Color codes score lower in maturity but higher in logical-gate efficiency because their transversal structure can reduce the cost of some logical operations [8], [16], [20]. If future superconducting and neutral-atom platforms continue to improve stabilizer extraction and decoding, color codes could become the better logic-oriented alternative for parts of a full quantum computer. Bosonic codes score asymmetrically: they are excellent when the hardware natively supports them but do not serve as a universal one-size-fits-all replacement [9], [18], [21]. qLDPC codes score highest in overhead reduction and long-term promise, but lowest in maturity and full-stack decoder-hardware readiness [13]-[15].

Year	Platform / team	Code emphasis	Representative result
2022	Krinner et al.	Surface code	Repeated distance-three surface-code cycles with 1.1 μs cycle time on superconducting hardware [10].
2023	Google Quantum AI	Surface code	Break-even style scaling from d = 3 to d = 5; larger code modestly outperformed smaller code [11].
2023	Miao et al.	Leakage handling	Leakage-removal protocol cut steady-state leakage by about tenfold and kept average leakage below 1 × 10^-3 [12].
2024	Bluvstein et al.	Logical processor	Up to 280 physical qubits and 48 logical qubits on reconfigurable atom arrays, including color-code

			milestones [16].
2024	Bravyi et al. / IBM	qLDPC memory	Reported ~0.8% threshold and markedly lower overhead for logical memory than surface-code baselines [15].
2024	Bausch et al.	Neural decoding	Transformer-based decoder beat previous decoders on real and realistic surface-code data [17].
2024	Lachance-Quirion et al.	GKP / bosonic	Autonomous correction of GKP states increased logical lifetime in a superconducting device [18].
2025	Google Quantum AI	Surface code	Below-threshold distance-seven memory with 101 qubits and logical suppression factor $\Lambda = 2.14 \pm 0.02$ [19].
2025	Lacroix et al.	Color code	Distance-three to distance-five color-code suppression and efficient logical operations on a superconducting processor [20].

Table 3. Representative milestones showing how the QEC frontier has moved from theory to experimentally differentiated architectures.

Decoder design deserves separate emphasis because it often decides whether a code family is merely elegant or operationally viable. Surface-code decoding has benefited from decades of work, from maximum-likelihood techniques [22] to almost-linear union-find-style approaches [23] and recent learned decoders such as AlphaQubit, which outperformed state-of-the-art methods on real-world data from Google's superconducting processor [17]. This decoder ecosystem is a hidden strategic advantage of the surface code: it reduces integration risk because both accuracy-focused and latency-focused options already exist.

Another way to compare code families is through the experimental milestones listed in Table 3. These milestones reveal a clear pattern. Surface-code experiments have repeatedly advanced memory scaling, leakage handling, and decoder integration [10]-[12], [19]. Neutral-atom experiments have demonstrated broad logical programmability and shown that color-code-style logic deserves serious attention [16]. Bosonic experiments are beginning to cross meaningful break-even thresholds in specialized hardware [18]. Meanwhile, qLDPC has moved from abstract promise to highly credible architecture proposals with explicit overhead advantages [15]. The frontier is therefore no longer a debate about whether QEC is possible; it is a debate about which stack will scale best under realistic engineering constraints.

In that sense, best depends on regime. For a near-term universal machine built from today's superconducting or similar planar qubits, the best current answer is surface code. For gate-efficient logical operations after modest hardware improvement, color code is the strongest challenger. For cavity-based or oscillator-rich hardware, bosonic and GKP encodings may be best at the first protection layer. For the long-term resource economy of memory-heavy architectures, qLDPC may become the eventual winner.

The proposal in the next section is therefore intentionally hybrid: it tries to choose the best current backbone without ignoring the future.

6. Proposed Best-Fit Current Architecture

The comparative evidence suggests that the most defensible answer to the question of best current QEC architecture is not 'surface code forever' and not 'qLDPC immediately.' The stronger answer is a staged hybrid stack. For near-term general-purpose computation, the computational backbone should remain the rotated surface code because it has the highest combination of maturity, locality, decoder availability, and repeated experimental validation. However, that backbone should be augmented with the best recent advances in leakage suppression, learned decoding, and modular logical operations. I refer to this proposed stack as the Adaptive Hybrid Surface Architecture (AHSA).

AHSA is best understood as an implementation strategy rather than a claim of a brand-new mathematical code. That distinction matters. Many student papers overstate novelty by proposing a new name for a system that has never been simulated or benchmarked. This paper instead proposes a best-fit architecture assembled from the strongest validated ideas in the literature. Its novelty is the synthesis and the design logic: which components should be combined now, which should remain optional, and which should be deferred until the hardware and decoder ecosystem mature.

Proposed architecture: Adaptive Hybrid Surface Architecture (AHSA)

1. Surface-code compute layer. Active computation should be organized around rotated surface-code patches because they fit planar hardware, support lattice-surgery communication, and already have a strong body of experimental evidence [6], [7], [10]-[12], [19].
2. Two-stage real-time decoder. The fast path should use a low-latency classical decoder such as union-find-style or matching-based decoding, while ambiguous or high-risk syndromes should be re-evaluated by a learned reranker or soft-information model trained on device-calibrated data [17], [23]. This preserves real-time operation while exploiting the accuracy gains of learned decoders.
3. Leakage-aware correction cycle. Leakage should be treated as a first-class design problem rather than an afterthought. The correction schedule should include periodic or continuous leakage removal inspired by the experimental results of Miao et al. because correlated leakage can destroy the assumptions of otherwise strong decoders [12].
4. Modular lattice-surgery operations. Logical interactions between patches should rely on lattice surgery rather than expensive nonlocal operations. This choice keeps the architecture compatible with planar layouts and aligns with the strongest resource-aware surface-code literature [7].
5. Optional bosonic front-end. On platforms with long-lived cavities or oscillator modes, a bosonic layer such as GKP can be used as a front-end to lower the effective error rate before information reaches the qubit-level topological code [18], [21]. This is optional rather than mandatory because not all hardware can support it.
6. Future qLDPC memory tier. AHSA leaves room for a migration path in which long-lived memory banks move from surface-code storage toward qLDPC storage as hardware connectivity, syndrome extraction, and decoder pipelines mature [13]-[15]. In that later stage, the surface code may remain the compute layer while qLDPC handles memory-heavy workloads.

The value of AHSA is that it chooses the most mature present backbone without locking the architecture into the highest-overhead long-term future. Pure qLDPC is not yet the safest choice because its full-stack implementation burden remains less tested than the surface code. Pure color-code architectures are

promising for logic but still carry more decoder and stabilizer-measurement risk in most present-day superconducting settings [20]. Pure bosonic architectures are highly attractive but platform-specific [18], [21]. Surface code plus intelligent decoding plus leakage control is therefore the strongest current default, while optional bosonic and future qLDPC layers give the design a credible evolution path.

A practical validation program for AHSA is straightforward enough for student-level simulation research. One can benchmark surface-code patches under circuit-level depolarizing noise, measurement bias, and injected leakage events; compare a baseline decoder against a learned reranker; and report metrics such as logical error per cycle, decoder latency, physical qubits per logical qubit, and success rate of lattice-surgery operations. A second stage can model a heterogeneous extension in which a bosonic front-end or qLDPC memory bank reduces overall overhead. Even if the full architecture is not experimentally realizable in one lab today, its components are concrete enough to support meaningful simulation and comparison work.

Metric	Why it matters	Desired trend in AHSA
Logical error per cycle	Primary measure of whether scaling actually improves the logical qubit.	Decrease with code distance
Decoder latency	Real-time correction fails if classical processing lags behind syndrome cycles.	Stay below cycle-budget target
Physical qubits per logical qubit	Captures the architecture's true resource burden.	Lower than pure surface baseline over time
Leakage population	High leakage breaks model assumptions and spreads correlated faults.	Remain bounded and actively suppressed
Lattice-surgery logical failure	Tests whether patch-to-patch computation remains practical.	Lower than idle-memory cost
Magic-state or non-Clifford cost	Universal computation remains bottlenecked by non-Clifford resources.	Reduce factory overhead per useful algorithm

Table 4. Recommended evaluation metrics for validating the proposed AHSA stack in future simulation or hardware studies.

Most importantly, AHSA defines best in the right way. It is the best present engineering choice, not the best asymptotic theorem. In quantum architecture, that distinction is everything.

7. Discussion, Limitations, and Future Work

The proposed architecture aligns closely with the direction of current industrial and academic research. Google's recent work emphasizes the surface code, leakage suppression, color-code logic, and learned decoding [11], [12], [17], [19], [20]. IBM's recent work emphasizes low-overhead qLDPC memories and hardware-code co-design [15]. Harvard-led neutral-atom research emphasizes logical control at scale and code-family flexibility [16]. AHSA can be read as a synthesis of these trends: use what is mature now, borrow what improves the stack today, and keep the door open to lower-overhead future layers.

At the same time, several unresolved problems prevent any final declaration of victory for one code family. First, cross-paper comparisons remain imperfect because thresholds and logical error rates depend heavily on noise assumptions, cycle design, post-selection rules, and whether leakage or crosstalk are

modeled explicitly. Second, the cost of universal computation is still dominated by non-Clifford resources, so a code with excellent memory behavior may still be unattractive if its T-gate pipeline is too expensive. Third, real-time decoding remains a systems challenge: the best decoder on paper is useless if it cannot keep up with microsecond-scale correction cycles [17], [19].

This study itself has limitations. It is literature-based and conceptual; it does not report original hardware experiments. Its scoring matrix is qualitative rather than a statistical meta-analysis, because the underlying papers do not all share identical metrics. Its proposal is therefore best interpreted as an evidence-based design recommendation rather than as a proven architecture. That honesty is a strength, not a weakness. The purpose of a first research paper is often to clarify the landscape, identify the strongest current direction, and propose a credible next step.

Future work should proceed in three layers. The first is comparative simulation: evaluate surface, color, bosonic-front-end, and qLDPC-enhanced stacks under unified noise models. The second is decoder research: quantify when learned reranking actually changes logical performance enough to justify added complexity. The third is architecture co-design: determine how modular links, chiplets, or neutral-atom transport can make qLDPC or heterogeneous codes practical without losing threshold advantages. These are exactly the kinds of questions that can turn a literature-based first paper into a simulation paper and eventually into a genuine original contribution.

8. Conclusion

Quantum error correction has matured from a foundational idea into the main engineering path toward large-scale quantum computing. The field now contains several serious code families, each with a different balance of threshold, overhead, locality, logical-gate cost, and hardware maturity. This paper compared the leading existing approaches and argued that no universal winner exists across all regimes. Concatenated codes remain foundational, color codes are increasingly compelling for logical operations, bosonic and GKP encodings are powerful on the right hardware, and qLDPC codes are the leading long-term route to lower overhead.

For near-term general-purpose fault-tolerant computation, however, the strongest present choice is still a surface-code-centric architecture. The evidence is straightforward: the surface code has the deepest experimental validation, the richest decoder ecosystem, strong locality, and repeated demonstrations of scaling, leakage handling, and below-threshold operation [10]-[12], [17], [19]. The best current proposal is therefore not to abandon the surface code, but to improve it intelligently. The Adaptive Hybrid Surface Architecture proposed here does exactly that by combining a surface-code compute layer with leakage-aware cycles, fast classical decoding, learned reranking, modular lattice surgery, and optional bosonic or future qLDPC layers.

In that sense, the best QEC strategy today is hybrid, staged, and hardware-aware. That is the conclusion most consistent with the current literature and the most useful starting point for future simulation, design, and experimental work.

Appendix A. Representative Literature Corpus

Appendix A lists twenty representative papers used to structure the comparison in this study. The list is intentionally balanced across foundational theory, code families, decoder research, and recent experiments.

Ref.	Year	Focus	Role in this study
[1]	1995	Shor code	Historical starting point for quantum redundancy and syndrome-based protection.
[2]	1996	Steane / CSS	Foundational stabilizer and CSS framework.
[3]	2003	Anyons / toric ideas	Topological basis for fault-tolerant coding.
[4]	2002	Topological memory	Threshold and geometric interpretation of surface-code behavior.
[5]	2015	QEC review	General background and evaluation framework.
[6]	2012	Surface code	Canonical practical surface-code architecture.
[7]	2012	Lattice surgery	Resource-aware logical operations for surface-code systems.
[8]	2006	Color code	Foundational topological color-code logic ideas.
[9]	2001	GKP code	Foundational bosonic / oscillator encoding.
[10]	2022	Repeated surface-code QEC	Important superconducting hardware milestone.
[11]	2023	Break-even scaling	Early evidence that larger surface codes can outperform smaller ones.
[12]	2023	Leakage suppression	Demonstrates why leakage must be a first-class design problem.
[13]	2021	qLDPC overview	Frames the promise and challenges of qLDPC codes.
[14]	2021	Balanced-product qLDPC	Explicit low-overhead qLDPC family.
[15]	2024	IBM qLDPC memory	Strong recent case for low-overhead fault-tolerant memory.
[16]	2024	Neutral-atom logical processor	Large logical processor with code-family flexibility.
[17]	2024	Neural decoding	State-of-the-art learned decoder for realistic surface-code data.
[18]	2024	Autonomous GKP QEC	Experimental bosonic milestone.
[19]	2025	Below-threshold surface code	Best current evidence for large-scale surface-code progression.
[20]	2025	Color code on	Latest major logic-oriented

		superconducting chip	color-code experiment.

Appendix Table A1. Twenty representative papers used to structure the comparison.

References

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